Cloud Computing

Exercise - 3 Total points: 10

Understanding Elastic Block Store and the cloud services in general

In this practice session, you will try to understand the importance of persistent storage with the help of the following exercise.

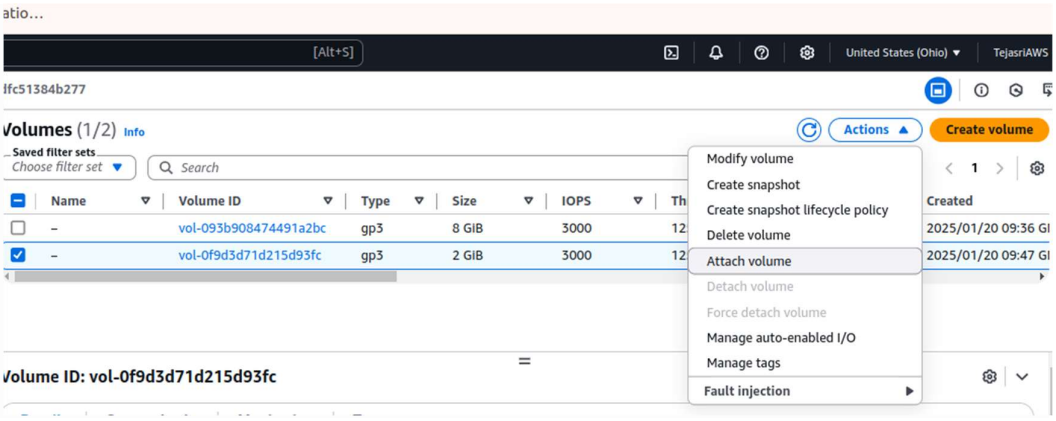
NB: Make sure you do this exercise with Amazon Machine Image being “Ubuntu” and not “Amazon Linux Image”.

Task 3.1: Attach a volume to an instance

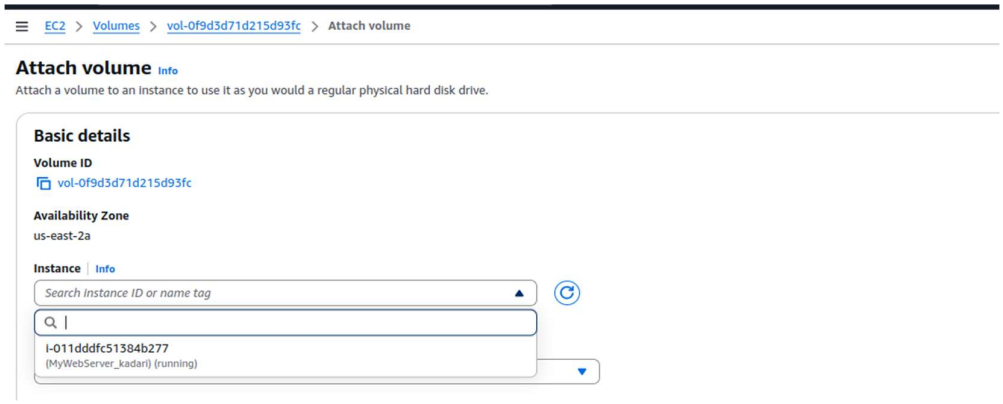
- First run an instance

Volumes are virtual hard disks, which can be used as a more permanent storage of files or extending the available disk space of an instance. We will create a new volume and attach it to our instance to increase the available disk space.

- Create a new 2GB size volume (EC2 DashBoard -> Volumes -> Create Volume)
  - Use your last name as the name of your volume.
- Attach the volume to your running instance.



- Then add the volume to one of the listed instances



- Then also add the device name "/dev/sdX" or whatever option in the drop down works without warning.

- Create a new disk partition on the attached volume
  - Log into your instance through ssh
  - Check the list of available disks and their partitions using the `lsblk` command line command
  - The name of the new disk added as a result of attaching your cloud volume might be `xvdb` or `sdb`
  - Create a new partition on the added disk
    - Run `sudo fdisk /dev/xvdb` command to start the partitioning process
    - Press `n` to create a new disk partition.
    - Press `p` to create a primary disk partition.
    - Press `1` to denote it as the 1st disk partition.
    - Press `ENTER` twice to accept the default of 1st and last cylinder – to convert the remainder of hard disk to a single disk partition
    - Press `t` to choose a type for the new partition.
    - Press `83` change your new partition to Linux partition type.
    - Finally, press `w` to commit changes.
- Mount the volume inside the instance to `/data` folder
  - Create the `/data` folder: `sudo mkdir /data`
  - Create a ext4 type filesystem on the new partition: `sudo mkfs.ext4 /dev/xvdb1`
  - Mount the partition under `/data` folder: `sudo mount -t ext4 /dev/xvdb1 /data`
- Create a new file in the mounted folder to verify that everything is working correctly.
  - You can use the `nano` command line text editor: `sudo nano /data/myfile.txt` (CTR+X keys to exit the editor)
- Run `lsblk` command in the command line interface inside your cloud instance and take a screenshot of the output. (2 Marks)

Task 3.2: Create a Web Server, with EBS Volume storing the Web Server files.

You should have observed in exercise 2, where you have created a web server that the web server instance when deleted/terminated has lost the modified web page.

This is the main reason we are using the persistent storage.

- Create an instance or use the instance from the previous task.
- Create a new volume like in the previous task and mount it at the path /var/www/html instead of /data. (**NB**: Make sure the mounting of the volume to path happens before apache2 web server software is installed)
- Now Install the web server like you have done in the previous exercise.
- Update the index.html file at /var/www/html path
- Now check your work by accessing the web page in the browser and look for your specific changes.

Take screenshots. 1. The web page screenshot should have your public IP/public DNS name of your instance from Amazon. 2. Should have your specific changes with your student ID visible. 3. Showing the available disks and their partitions. 4. Get a screenshot of Volume Summary at Volumes page at EC2 DashBoard for the volume attached, which should include the AWS instance id that it is attached to (2 Marks)

Task 3.3: Checking the persistent storage in action.

1. Now delete the instance created in Task 3.2
2. Create a new instance and then attach the EBS from the previous task to this running instance. (**NB**: you already have the partitioned the volume, no need to partition or format the volume again)
3. Then run
   ```
   sudo apt update
   ```
   ```
   sudo apt install apache2
   ```
4. Now mount the volume at the path /var/www/html, which would restore your modified web server files.
   ```
   mount /dev/xvdbb /var/www/html
   ```
5. Check if the modified index.html is restored with the new instance and old volume

Take screenshots: 1. Get the screenshot of Volume Summary. NB. The instance id to which it is attached must change, while volume id remains the same. 2. Screenshot of mount. 3. Screenshot of the webpage. (1 Mark)

You must delete the following after finishing your exercise

1. Your AWS Instances
2. Your volumes that were created.

Task 3.4 –

Choose ONE application that you personally use regularly

(Google Drive, Gmail, Google Classroom, Moodle, Microsoft OneDrive, Dropbox, Overleaf, Coursera, NPTEL, Udemy, GitHub, GitLab, Bitbucket, Replit, CodeSandbox, Stack Overflow, Docker Hub,

Postman, Google Scholar, ResearchGate, Zotero, Mendeley, Notion, Evernote, Todoist, Google Calendar, Microsoft Teams, Slack, Zoom, Google Meet, YouTube, Netflix, Amazon Prime Video, Amazon Kindle, Audible, Goodreads, Amazon, Flipkart, Myntra, Swiggy, Zomato, Blinkit, Zepto, Uber, Ola, Rapido, Google Maps, IRCTC Rail Connect, RedBus, Google Pay, PhonePe, Paytm, BHIM UPI, Splitwise, LinkedIn, LinkedIn Learning, Internshala, Unstop, Spotify, Hotstar, JioCinema, Telegram, Discord)

All the applications above are SaaS for the end users, We want you to decode what IaaS and PaaS might exist beneath them. We wouldn't expect exact services, but your assumptions of what services might exist.

1. Application Overview (0.5 point)

Describe the functionality of the application.

- What problem does it solve?
- What are the main features offered to users?

2. Application Requirements (1 point)

2.1 Computational Requirements

- What type of computation does the application require?

(e.g., request handling, data processing, real-time processing, background jobs)

2.2 Storage Requirements

- What kind of data is stored? (e.g., user data, files, logs, transaction records)
- Is the data read-heavy, write-heavy, or both?

2.3 Users

- Who are the users of the application?
- Are they end users, administrators, service providers, or all of these?

2.4 Functional and Non-Functional Requirements

- List two functional requirements
- List two non-functional requirements

3. Elasticity and Cost Efficiency (0.5 point)

- When do you foresee the need for elasticity in this application?
- During such periods, which AWS services (IaaS or PaaS) would be more cost-efficient?
- Explain why.

4. Cloud Benefits (0.5 point)

What advantages does the application gain by being deployed on the cloud instead of a traditional on-premise infrastructure?

5. Final Scenario-Based Question: IaaS vs PaaS with AWS Mapping

Consider a SaaS application of your choice

Assume the application is initially deployed using an IaaS model on AWS.
Later, the same application is migrated to a PaaS model on AWS.

Answer the following questions:

a. IaaS Deployment (AWS – Infrastructure Level) (0.5 point)

If the application is deployed using an IaaS model, explain:

- What responsibilities the system architect / developer must handle
- Which AWS services would be used for:
    - Compute
    - Storage
    - Database
    - Networking
    - Monitoring and scaling

b. PaaS Deployment (AWS – Platform Level) (1 point)

If the same application is deployed using a PaaS model, explain:

- Which responsibilities are now managed by AWS
- What responsibilities still remain with the developer
- Which AWS PaaS services would be used to deploy the application

c. Migration Analysis: IaaS → PaaS (1 point)

When migrating the application from IaaS to PaaS, discuss:

- Two advantages of moving to PaaS
- Two disadvantages or limitations of moving to PaaS

Follow these guidelines while submitting the exercises

1. Generate a PDF file with answers/screenshots with name being your studentid_EX3
2. Put a pdf file with the answers for the questions
    a. For programming tasks show screenshots of the program output for the correct and wrong inputs
    b. For commands tried, provide the screenshots
3. Do not copy code/answers from internet. If you copy, please leave a comment with the URL on the file
4. Plagiarism will not be tolerated and dealt through the formal channels of the university